Problem Statement

Due to lack of central platform for patient's medical records there are so many difficulties in analyzing patient's health conditions in different hospitals and lack of adapting recent technologies like IoT, AI and Blockchain are becoming barrier for Smart Health Care.

Proposed Solution

Using IoT technology to record patient's health parameters on real time when patient admitted in hospital and preparing the health record of that incident and storing in blockchain for future health problems. Patient can see his/her full medical history at single central platform and can share with any doctor or hospital for any medical treatment. The medical history of patient can be shared researchers to research on critical diseases diagnosis using AI technology. The medical bills also can be utilized to settle in case insurance claims using blockchain transactions.

Brief Description

Smart Health Care is not only about patient monitoring at one medical problem, the medical history recorded using technology should be utilized to diagnosis future medical problems. Digitizing the medical reports on a centralized platform will make the hospitals paperless, which will make the process easy for patients as well as doctors.

We aim to create an IoT device with the sensors such as Heartbeat, blood pressure, body temperature etc. for collection of patient's real time health parameters when he/she is admitted in hospital or home.

Using blockchain technology we aim to store the patient's medical history in a centralized platform. User can share his/her medical records with any doctor or hospital. With this approach there is no need of paper records to be carried every time when there is any health problem.

By sharing the medical records with researchers, they can use AI technology to diagnosis the best medicine to health problems.

Sample use case as below:

1) Imagine any patient is having health problem and admitted in a hospital.
2) Hospital will use blockchain based centralized platform to admit the patient (registration)
3) Patient will share his/her medical records with the hospital
4) Doctor will use the records shared by the patient and analyze the treatment required
5) IoT device will be attached to patient while he is bed to monitor the real time health parameters
6) IoT device will send regular updates to doctors and nurses through mobile notifications
7) Doctor can see the patient's real time health parameters on cloud platform
8) At the end of each day nurse will push the medicine used for the treatment and the health parameters after medication to the blockchain
9) At the end of treatment centralized platform will generate the medical record of that incident and securely stores in blockchain
10) If the same patient admits in different hospital for some health problem, he can share his previous medical records to the new hospital through blockchain central platform

The above use case is also applicable when patient is admitted at home. Hospital will continuously monitor patient health parameters from the IoT device attached at home.

A smart hospital needs to look beyond its walls with a population health view to get patients discharged quicker while ensuring they recover well, without being required to be readmitted.

This centralized platform and IoT device can be utilized by all the hospitals for smart treatment. Readily data available for researchers to improve medical treatment.

Technologies

IoT – For collection of real time health parameters, Blockchain (Hyperledger) – Storage of medical records and transactions, AI – For research on critical diseases diagnosis, Cloud Platform – AWS and/or IBM for running the applications, AngularJS – For better User Experience, MongoDB – For temporary storage of IoT health parameters, RaspberryPi – For interaction of all the sensors.